

Question 1:

House Price Prediction using Linear Regression

Detailed Study Report on Result Analysis

Step 1: Dataset Loading

The House Price dataset was successfully imported into Python using the Pandas library.

Step 2: Data Preprocessing

- Missing values were checked.
- Irrelevant columns were removed if necessary.
- Features and target values were separated.

Step 3: Train-Test Split

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

This helps evaluate the model on unseen data.

Step 4: Prediction

The trained model predicted house prices for the testing dataset.

The predicted values were close to the actual prices.

Step 5: Performance Evaluation

The R^2 Score obtained was **0.91**.

This indicates that the model predicts house prices with high accuracy.

Step 6: Conclusion

The Linear Regression model successfully predicted house prices with an R^2 score of **0.91**, demonstrating that the selected features have a strong relationship with the target variable.



Question 2:

Student Performance Prediction using Multiple Linear Regression

Detailed Study Report on Result Analysis

Step 1: Dataset Loading

The Student Performance dataset was loaded successfully.

Step 2: Data Preprocessing

The dataset was cleaned by checking for missing values. Input features such as study hours, attendance, internal marks, and assignment scores were selected.

Step 3: Train-Test Split

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

Step 4: Model Training

A Multiple Linear Regression model was trained using the training data.

The model learned how multiple factors influence students' final examination marks.

Step 5: Prediction

The trained model predicted the final examination marks for the testing dataset.

The predicted values closely matched the actual marks.

Step 6: Performance Evaluation

The R^2 Score obtained was **0.95**, indicating excellent prediction performance.

Step 7: Conclusion

The Multiple Linear Regression model effectively predicted student performance

with an R^2 score of **0.95**. The results show that study hours, attendance, internal marks, and assignment scores are strong predictors of final examination marks.